Tangent & Normal: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Tangent and normal at a point

Formulas

$$m = \left.\frac{dy}{dx}\right|_{(x_1,y_1)}, \quad T:\ y - y_1 = m(x - x_1), \quad N:\ y - y_1 = -\frac{1}{m}(x - x_1)$$

KEY POINTS

- Tangent = limiting position of secant PQ as $Q \to P$; normal $\perp$ tangent at P
- $m = 0$: $T:\ y = y_1$, $N:\ x = x_1$; $m \to \infty$: $T:\ x = x_1$, $N:\ y = y_1$
- Parametric: $m = \frac{dy/dt}{dx/dt}$ at t_0, point $(x(t_0), y(t_0))$
- Implicit curve: differentiate implicitly, then substitute the point
- Piecewise / modulus: at a junction need continuity and equal finite $f'(a^-) = f'(a^+)$; else no unique tangent
- A tangent can cut the curve again, even at the point of contact ($y = x^3$ at O); a line is its own tangent

Watch out

Find y_1 from the curve before writing the line; m is the value at the point, not $\frac{dy}{dx}$ in general

2. Quick objective facts

KEY POINTS

- Second-degree curve: tangent at (x_1, y_1) by $x^2 \to xx_1$, $y^2 \to yy_1$, $2xy \to xy_1 + yx_1$, $2x \to x + x_1$, $2y \to y + y_1$
- Even function: $f'(0) = 0$ if it exists; tangent at $x = 0$ is $y = f(0)$
- Polynomial curve through O: tangent(s) at O = lowest-degree terms $= 0$
- Equal non-zero intercepts: slope -1; intercepts equal in magnitude: slope ± 1
- Tangent parallel to x-axis: $\frac{dy}{dx} = 0$; parallel to y-axis: $\frac{dx}{dy} = 0$

3. Vertical tangent

KEY POINTS

- f continuous at x_0; $f'(x_0^-)$, $f'(x_0^+)$ both $+\infty$ or both $-\infty$: vertical tangent $x = x_0$
- One $+\infty$, other $-\infty$ (cusp): no vertical tangent
- $x^{1/3}$, $\operatorname{sgn} x$: vertical tangent at 0; $x^{2/3}$, $|x|$: none

Watch out

Vertical tangent means f is not differentiable there, yet a tangent exists

4. Parametric points

Formulas

$$(a\cos\theta, a\sin\theta),\ (a\cos\theta, b\sin\theta),\ (a\sec\theta, b\tan\theta),\ \left(ct, \frac{c}{t}\right),\ (at^2, 2at),\ (2at, at^2)$$

$$\frac{x^n}{a^n} + \frac{y^n}{b^n} = 1:\ \left(a\cos^{2/n}\theta,\ b\sin^{2/n}\theta\right)$$

KEY POINTS

- Order: circle, ellipse, hyperbola, $xy = c^2$, $y^2 = 4ax$, $x^2 = 4ay$
- $y = f(x)$: $(t, f(t))$; line $y = mx + c$: $(t, mt + c)$
- Astroid $x^{2/3} + y^{2/3} = a^{2/3}$: $(a\cos^3\theta, a\sin^3\theta)$, slope $-\tan\theta$; tangent cuts a constant length a between the axes

5. Ultimate concept: tangent / normal through a point, common tangent

Formulas

$$f'(t) = \frac{f(t) - y_1}{t - x_1} \quad \text{(tangent)}, \qquad -\frac{1}{f'(t)} = \frac{f(t) - y_1}{t - x_1} \quad \text{(normal)}$$

$$f'(t_1) = g'(t_2) = \frac{g(t_2) - f(t_1)}{t_2 - t_1}$$

STEPS

1. Take the unknown point of contact $P(t, f(t))$
2. Equate slope of tangent (or normal) at P to slope of P and the given point
3. Solve for t; each real t gives one line

KEY POINTS

- Tangent at P meeting the curve again at Q: same equation with $Q(t, f(t))$ and $m = f'(P)$; cancel the factor $(t - x_P)$
- $y = x^3$: tangent at $x = a$ meets again at $x = -2a$; slope there is 4 times
- Line tangent at P and normal at Q: $m_T(P) = m_{PQ} = -\frac{1}{m_T(Q)}$

Watch out

Never use $f'(x_1)$ of the external point: it is not on the curve

6. Minimum distance

KEY POINTS

- Point A to curve: AP extreme when AP is normal at P
- Two curves: 0 if they meet; else along the common normal
- Curve and line: tangent at P parallel to the line, $d =$ distance of P from the line
- Curve and circle: normal at P passes through centre C, $d = CP - r$
- Circle touching a curve at P: its centre lies on the normal at P
- $(x_1 - x_2)^2 + (y_1 - y_2)^2$ with each point on a curve = squared distance between the curves

Watch out

Check the found point is the nearest one (other roots can give maxima or a far branch)

7. Angle of intersection and orthogonal curves

Formulas

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|, \qquad \perp:\ m_1 m_2 = -1$$

$$a_1 x^2 + b_1 y^2 = 1 \ \perp\ a_2 x^2 + b_2 y^2 = 1 \iff \frac{1}{a_1} - \frac{1}{b_1} = \frac{1}{a_2} - \frac{1}{b_2}$$

STEPS

1. Solve the curves together for the common point(s)
2. Slopes m_1, m_2 of both curves at that point
3. Acute angle from $\tan\theta$; one slope infinite: $\tan\theta = \left|\frac{1}{m}\right|$

KEY POINTS

- No common point: angle not defined; several points: find the angle at each
- Orthogonality without the point: show $m_1 m_2 = -1$ using both equations
- $xy = a^2$ and $x^2 - y^2 = b^2$ always orthogonal; confocal conics cut orthogonally

8. Curves touching each other; number of solutions

Formulas

$$f(x_1) = g(x_1), \qquad f'(x_1) = g'(x_1)$$

KEY POINTS

- Touch the x-axis at a: $f(a) = 0,\ f'(a) = 0$
- $e^x = kx$: touch $k = e$; $k < 0$: 1, $0 \le k < e$: 0, $k = e$: 1, $k > e$: 2
- $\ln x = kx$: touch $k = \frac{1}{e}$; $k \le 0$: 1, $0 < k < \frac{1}{e}$: 2, $k = \frac{1}{e}$: 1, $k > \frac{1}{e}$: 0
- $\ln x = kx^2$: touch $k = \frac{1}{2e}$; $k \le 0$: 1, $0 < k < \frac{1}{2e}$: 2, $k = \frac{1}{2e}$: 1, $k > \frac{1}{2e}$: 0
- $a^x = x$ $(a > 1)$ has a root iff $a \le e^{1/e}$ (touch at $x = e$)

STEPS

1. Write the equation as curve = line / simple curve through a fixed point
2. Find the touching parameter (equal values and equal slopes)
3. Rotate about it and count intersections in each range

9. Important lengths

Formulas

$$PT = \left|\frac{y_1\sqrt{1+m^2}}{m}\right|,\ PG = \left|y_1\sqrt{1+m^2}\right|,\ TM = \left|\frac{y_1}{m}\right|,\ MG = |y_1 m|$$

$$T\left(x_1 - \frac{y_1}{m}, 0\right), \qquad G(x_1 + my_1, 0)$$

KEY POINTS

- T, G = where tangent and normal meet the x-axis; M = foot of ordinate
- $y = be^{x/a}$: subtangent = $|a|$ constant; $y^2 = 4ax$: subnormal = $2a$ constant
- Proof: $\tan\psi = m$; $PT = y\operatorname{cosec}\psi$, $TM = y\cot\psi$, $PG = y\sec\psi$, $MG = y\tan\psi$

Watch out

Lengths are positive: keep the modulus

10. Approximation by differentials

Formulas

$$f(x + \Delta x) \approx f(x) + f'(x)\,\Delta x, \qquad dy = f'(x)\,dx$$

KEY POINTS

- Pick x near the number with exact $f(x)$; Δx small
- Geometry: the curve is replaced by its tangent; $dy \approx \Delta y$
- $f'' > 0$: approximation below actual; $f'' < 0$: above actual
- Relative error $\frac{dy}{y}$; percentage error $\frac{dy}{y} \times 100$

11. Rate measure (related rates)

Formulas

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

STEPS

1. Draw the figure and name the changing quantities
2. Write a relation true at every instant (similar triangles, Pythagoras, volume)
3. Differentiate w.r.t. t, then substitute the values of that instant

KEY POINTS

- Decreasing quantity has negative rate
- Cone with semi-vertical angle α: $r = h\tan\alpha$, $V = \frac{1}{3}\pi h^3\tan^2\alpha$
- Sphere: $\frac{dV}{dt} = 4\pi r^2\frac{dr}{dt}$, $\frac{dS}{dt} = 8\pi r\frac{dr}{dt}$
- On a curve: y changes k times as fast as x when $\frac{dy}{dx} = k$; x faster than y when $\left|\frac{dy}{dx}\right| < 1$

Watch out

Substitute instantaneous values only after differentiating

12. Common mistakes checklist

STEPS

1. Point really on the curve? Find the missing coordinate first
2. Normal slope is $-\frac{1}{m}$, not $\frac{1}{m}$
3. Parametric slope is $\frac{dy/dt}{dx/dt}$, not $\frac{dx/dt}{dy/dt}$
4. External point: use $P(t, f(t))$, never f' at the external point
5. Angle of intersection is acute; check each common point
6. Touching needs equal values and equal slopes
7. Keep modulus in lengths and distances